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Unit – 1 : Session – 10 : SLO – 1 | Logic Programming Simulation

■ Activity (Solutions)

- 1) Expert-system rule in Java: `int age = 20; if (age > 18) System.out.println("Eligible to vote"); else System.out.println("Not eligible");` — mimics the Prolog rule `eligible(X) :- age(X,A), A > 18.`
- 2) Decision rules → conditions: rule 'discount if member AND purchase > 1000' → `boolean discount = isMember && amount > 1000;` rule 'alarm if temp > 40 or smoke detected' → `if (temp > 40 || smoke) alarm();`
- 3) Rule-based quiz app design: `class Quiz { String question; Map<Character,String> options; char key; boolean check(char ans){ return ans == key; } }` — each question object stores its rule (correct key); the engine asks, reads input, applies the rule `check()`, and scores: `score += q.check(ans) ? 1 : 0.`